cenit-II

Requirement Analysis

Requirement Engineering Process:

To gather stequirements from customer, analyze and document them.

Steps:

انا

1) Requirement Elicitation & Analysis
2) software Requirement Specification

"

Validation

4

1/

Manage
ment

Stix

Feasibility
study

1.

Requirements
Elicitation 2

pri

Analysi
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feasibi
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Requirem
ent

specificati
on

boatza

Syste
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Requirem
ent

models

الماان

validation Dall

plitom

Wen &
System,

Puer bas

requireme
nts

Requireme
nts
documento

Stap 1.

x

Process of gathering actual requirements

Requirement Elicitation &

Elicitation & Analysis

Understands customer, busines manuals,

existing software of same type.

Tannique s :

CInterviews

C

Surveys

3 Buotot yping (creating

work model>

& Fous

Groups

4 observation

Step2: Software

Requirement specification

* Went are written in natural lang

+ requirements in tachnical

lang

Defin

es

* Data fece diagram, dat a

dict, ER diagram

O software will interact

with hlw

ง

External interface

Cspeed,

respond time

@ maint an

ability

Step 3. Software

Requirement *validation*

* documents are validated on

tested Requirement

reviews taken by

customers. After

deudoping prototyping

check with customers & Test

Case

generatio

n.

Conditions

practically

implement
any
ambiguities
* can ales vube

Step 4: Software
Requirement management *
managing changing
requirements

Activities:

1) Tracking 2) Version
url 3) Traceability.
Communication

Developing use cases & we
care model.

Use case model is *a* us visual
representation of how users.

interact with the system to
achieve specific
goals.

0

ussct it It helps in identifying the *systems* frenctional requirements from the users perspective.

the system.

and

It shouls who uses

uses

- what they do (mecare)
  how the system
  responds

Expa
ts

Pwrpare

Purpose of me care moder

&

It helps in underst

user requirement

clearly.

It act as a

bridge

between

requirement

gathering

and

It is used dur

requirement analysis

phone

-andi
ng

desig
n.

during

.

Easy for clients
and
non technical state
holders to understand.
us Components of a we
care diagram,
(1)
Actors

Represent
user/external sys
that interact with the system.
They are debicted
as
(stic
k)
stick
figures.
श_
(ii) use
cases:
action
Represents

especific

or task that actors can.

within the
system

(iii) System Boundary

f

✓ System Boundary is a box that visually represent the *system* enclosing all the use cases.

What care Benefits of we case Diagram?

use case Diagram offers a clear, visual representation of system

function and its
interaction with external
users.

a

ше адем We care dgrm
illatrate

the different
ways with the
system

users

eng
age
d

use Care Diagram Example

onine shopping system

Actors:

customer

3 Admin L

use cares:

→ Brouve Broducts

- Add to cart

→ checkout

register
ed

customer

View items

2

Authentication

Identity

petovider

make pumhare

Checkout

credit

шев

customer

Client

register

Payment

Service

new customer

the care Diagram for online Shopping System

Я

paypal

Pay Pal

I wont come.

Registe
r

shop

Login

Duner

Customer
сиют

Apditha

podanum

(Se auch

item)

Select *on* add

Payment

manag

er

item

make

payment

Order

history

notificati

ons

Stock qui

Library

management:

<u>Registe</u>

<u>D</u>

Togen

Shop owner

ɔ

admin

cus fomer

Search book

Jack

choose one

Or more book

úbravian

Bes borrow die, data

chik Stok

Now el

Alm machine

customer

сино

Syst
om

Alm maching

login

withdraw
al

dobit

Enter Pin

dalabare

メーメ

Die ungab

- Technical representation of system,

*

<u>Analysi</u>

S:

the

design

model

+ link between

syste

m

description

and

Analysis

modelling

*

»

informati
on

behaviour

hanslated into
component

auchitecture

intoyour level
design.

objectives of malyris
modelling.

* Understanding
Needs

* Communication

& process of analy
is modelling
helps in

understanding.

extraction of mer needs

* Analysis models

faùutat os

Communication between mers, clients,

developers and test.ors.

– clarifying

Ambiquities:

Analy sis

models

eusist in resolving requirements disputes and providing clarification

on unclear que as.

- finding data requirements:

e

1

Determining relations, entitles & qualities.

*

Montre

Refining

Defining

behavious:

Why the

Anely in

modeling

aids in definition of the

system

dynami

c

behaviour.

× System boundary

I donnipeater

It is made easier

by analysis modelling, **which**

helps in defining

paramotors of software

sypi

ant

other

Elements

Dolgo

Dala

and its interactions with wers,

system

s:

description

ER

Dala

Diagram

Didimary

1 Dara Dictionary:

Proces

lyficul

tor

Statel

Transition

Diagram

control

pals

flow

deuen

specificatio
n

Description of all
data obj

→ stares collection

of data

7-

→ Crucial dement

of

analysis

~2) ER
Diagram:

السبيل

Depicts relationship
between

data objects and is med in

conduring

data moa

model

ing.

Provides bauis

for activity

related to "data

services.

Negotiation

ulorking

when

People when

on a agreement to

solve prblms I make decision

together.

For eg. A customer might

want a

lot of features but the developers

don't have enough

know they do everything

.

time to

so the software

development discus

what is the most

on which

features

team important

and

agre

e

to do

first.

sienario Bard modelling /

Element

Duvity

Diagram

initial state

action *box*

Deijon box

Final stats

claw Band Modeling

D Claus Diagram

Clouswame

attributes

methods

Flow Oriented modelling

Element

@Data Flow

Diagram

entry!

Data Flow

Proces
s

Data slove

<

2 Contral Flow Diagram

Stout

Age=1
8

Eugibl
e

N7=8

NoteUsing

stop

Behavioral modelling /

Element

O →

Initialstate

O

→ Final *state*

→

simple state

Composite

state.

Types *of*

Pala moder

Desevent

ual

Drogi

cal

15.
Phypi
cal

Carry
al

-
high
level
representation
8

data and its
gelationship

- Physical

storage

conceptual

design phaso

Conceptual data

model:

-Entities

(obi)

•Student Winds Courie

•Teacher

•Attributes

(properties)

StudantD

Locane ID

Relationships

:

Student envolls in a

Courre

Logical data: [pdation

al Form

Tables:

refers to the stuctore of dato It will

be implemented in db

Builds upon conceptual

data model

Student

Cours e

Teacher

Dato

Types.

name

Age

Constraint
s:

Student-**ID**
primary key

Physical data
model:

on hardware .

acured..

.

Hierarchical
model:

Refers to actual
storage of data
Doals with how data
stored,

free like structure.

Stored as

records

-parent-child

elationship.